Exercise 3 — Step-by-Step Solutions

Advanced System Theory · Response of Linear Systems

Course: Advanced System Theory (Paderborn University, Automatic Control Group RAT, Prof. Dr. Erdal Kayacan) **Covers:** Task 1 (a–c), Task 2, Task 3, Task 4 (a–b) **Checking:** no official solution was available when this was written. Every result is verified by the scripts in `python/` — symbolic algebra (SymPy: comparison with the exact matrix exponential) and numerical simulation (SciPy).

Every step is written out. References like "Concepts §7.4" point to `concepts.md`. Notation as in the lecture: bold symbols are vectors/matrices, $\boldsymbol{\Phi}(t, t_0)$ is the state transition matrix, $\boldsymbol{P} = [\boldsymbol{p}_1\ \boldsymbol{p}_2]$ the eigenvector matrix with $\tilde{\boldsymbol{p}}_i$ the rows of $\boldsymbol{P}^{-1}$, $\alpha_k(t)$ the Cayley–Hamilton coefficients, $\sigma(t)$ the unit step, $\delta(t)$ the Dirac impulse, $j = \sqrt{-1}$.

Contents

- Task 1 — Modes and responses of LTI systems
- Task 2 — Recovering the state transition matrix
- Task 3 — Determining initial conditions
- Task 4 — Cayley–Hamilton method and response of LTI systems
- Summary of results

Task 1 — Modes and responses of LTI systems

$$\dot{\boldsymbol{x}}(t) = \underbrace{\begin{bmatrix} 0 & -2 \\ 1 & -1 \end{bmatrix}}_{\boldsymbol{A}} \boldsymbol{x}(t) + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\boldsymbol{B}} u(t), \qquad y(t) = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\boldsymbol{C}} \boldsymbol{x}(t), \qquad \boldsymbol{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

1(a) — State transition matrix by diagonalization

We use $e^{\boldsymbol{A}t} = \boldsymbol{P} e^{\boldsymbol{\Lambda} t} \boldsymbol{P}^{-1}$ (Concepts §7.1–7.2).

Step 1 — Eigenvalues.

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \det \begin{bmatrix} \lambda & 2 \\ -1 & \lambda + 1 \end{bmatrix} = \lambda(\lambda + 1) - (2)(-1) = \lambda^2 + \lambda + 2 = 0.$$

The discriminant $1 - 8 = -7$ is negative, so the roots are complex (Concepts §2.1):

$$\lambda_{1,2} = \frac{-1 \pm \sqrt{-7}}{2} = -\frac{1}{2} \pm j\frac{\sqrt{7}}{2}, \qquad \sigma = -\frac{1}{2}, \quad \omega = \frac{\sqrt{7}}{2} \approx 1.32.$$

Check: $\lambda_1+\lambda_2=-1=\operatorname{tr}\boldsymbol{A}$ ✓, $\lambda_1\lambda_2=\frac{1}{4}+\frac{7}{4}=2=\det\boldsymbol{A}$ ✓.

Step 2 — Eigenvectors. For $\lambda_1=-\frac{1}{2}+j\frac{\sqrt{7}}{2}$:

$$\lambda_1\boldsymbol{I}-\boldsymbol{A}=\begin{bmatrix}\lambda_1 & 2\\ -1 & \lambda_1+1\end{bmatrix}.$$

The matrix is singular, so the second row suffices: $-p_1+(\lambda_1+1)p_2=0$, i.e. $p_1=(\lambda_1+1)p_2$. With $p_2=2$ (to avoid fractions) and $\lambda_1+1=\frac{1}{2}+j\frac{\sqrt{7}}{2}$:

$$\boldsymbol{p}_1=\begin{bmatrix}1+j\sqrt{7}\\ 2\end{bmatrix},\qquad \boldsymbol{p}_2=\bar{\boldsymbol{p}}_1=\begin{bmatrix}1-j\sqrt{7}\\ 2\end{bmatrix}\quad(\text{for }\lambda_2=\bar{\lambda}_1).$$

Check: $\boldsymbol{A}\boldsymbol{p}_1=\begin{bmatrix}-4\\ (1+j\sqrt{7})-2\end{bmatrix}=\begin{bmatrix}-4\\ -1+j\sqrt{7}\end{bmatrix}$ and $\lambda_1\boldsymbol{p}_1=\frac{-1+j\sqrt{7}}{2}\begin{bmatrix}1+j\sqrt{7}\\ 2\end{bmatrix}=\begin{bmatrix}\frac{(-1+j\sqrt{7})(1+j\sqrt{7})}{2}\\ -1+j\sqrt{7}\end{bmatrix}=\begin{bmatrix}\frac{-1-7}{2}\\ -1+j\sqrt{7}\end{bmatrix}=\begin{bmatrix}-4\\ -1+j\sqrt{7}\end{bmatrix}$ ✓.

Step 3 — $\boldsymbol{P}$, $\boldsymbol{\Lambda}$ and $\boldsymbol{P}^{-1}$.

$$\boldsymbol{P}=\begin{bmatrix}1+j\sqrt{7} & 1-j\sqrt{7}\\ 2 & 2\end{bmatrix},\qquad \boldsymbol{\Lambda}=\begin{bmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{bmatrix},\qquad \det\boldsymbol{P}=2(1+j\sqrt{7})-2(1-j\sqrt{7})=4j\sqrt{7}.$$

With the 2×2 inverse formula and $\frac{1}{4j\sqrt{7}}=\frac{-j}{4\sqrt{7}}$:

$$\boldsymbol{P}^{-1}=\frac{1}{4j\sqrt{7}}\begin{bmatrix}2 & -(1-j\sqrt{7})\\ -2 & 1+j\sqrt{7}\end{bmatrix}=\begin{bmatrix}\tilde{\boldsymbol{p}}_1\\ \tilde{\boldsymbol{p}}_2\end{bmatrix},\qquad \tilde{\boldsymbol{p}}_1=\begin{bmatrix}-\dfrac{j}{2\sqrt{7}} & \dfrac{1}{4}+\dfrac{j}{4\sqrt{7}}\end{bmatrix},\quad \tilde{\boldsymbol{p}}_2=\bar{\tilde{\boldsymbol{p}}}_1.$$

($\tilde{\boldsymbol{p}}_1$: $\frac{-j}{4\sqrt{7}}\cdot 2=-\frac{j}{2\sqrt{7}}$ and $\frac{-j}{4\sqrt{7}}\cdot(-1+j\sqrt{7})=\frac{j+\sqrt{7}}{4\sqrt{7}}=\frac{1}{4}+\frac{j}{4\sqrt{7}}$.) *Check:* $\tilde{\boldsymbol{p}}_1\boldsymbol{p}_1=-\frac{j(1+j\sqrt{7})}{2\sqrt{7}}+2\left(\frac{1}{4}+\frac{j}{4\sqrt{7}}\right)=\frac{-j+\sqrt{7}}{2\sqrt{7}}+\frac{1}{2}+\frac{j}{2\sqrt{7}}=1$ ✓.

Step 4 — Multiply $\boldsymbol{P}e^{\boldsymbol{\Lambda}t}\boldsymbol{P}^{-1}$. Column times row (Concepts §8.1):

$$e^{\boldsymbol{A}t}=\boldsymbol{P}\begin{bmatrix}e^{\lambda_1 t} & 0\\ 0 & e^{\lambda_2 t}\end{bmatrix}\boldsymbol{P}^{-1}=\boldsymbol{p}_1\tilde{\boldsymbol{p}}_1e^{\lambda_1 t}+\boldsymbol{p}_2\tilde{\boldsymbol{p}}_2e^{\lambda_2 t}.$$

The first mode matrix:

$$\boldsymbol{M}_1=\boldsymbol{p}_1\tilde{\boldsymbol{p}}_1=\begin{bmatrix}(1+j\sqrt{7})\left(-\frac{j}{2\sqrt{7}}\right) & (1+j\sqrt{7})\left(\frac{1}{4}+\frac{j}{4\sqrt{7}}\right)\\ 2\left(-\frac{j}{2\sqrt{7}}\right) & 2\left(\frac{1}{4}+\frac{j}{4\sqrt{7}}\right)\end{bmatrix}=\begin{bmatrix}\frac{1}{2}-\frac{j}{2\sqrt{7}} & \frac{2j}{\sqrt{7}}\\ -\frac{j}{\sqrt{7}} & \frac{1}{2}+\frac{j}{2\sqrt{7}}\end{bmatrix}=\frac{1}{2}\Big(\boldsymbol{I}+\frac{j}{\sqrt{7}}\boldsymbol{N}\Big),$$

(Top-left: $-\frac{j}{2\sqrt{7}}-\frac{j^2\sqrt{7}}{2\sqrt{7}}=\frac{1}{2}-\frac{j}{2\sqrt{7}}$. Top-right: $\frac{1}{4}+\frac{j}{4\sqrt{7}}+\frac{j\sqrt{7}}{4}+\frac{j^2}{4}=\frac{j}{4\sqrt{7}}+\frac{7j}{4\sqrt{7}}=\frac{2j}{\sqrt{7}}$.) Since everything in the second mode is the complex conjugate, $\boldsymbol{M}_2=\bar{\boldsymbol{M}}_1$, and the sum of a number and its conjugate is twice its real part:

$$e^{\boldsymbol{A}t} = \boldsymbol{M}_1 e^{\lambda_1 t} + \bar{\boldsymbol{M}}_1 e^{\bar{\lambda}_1 t} = 2\,\mathrm{Re}\left[\boldsymbol{M}_1 e^{\lambda_1 t}\right].$$

With Euler, $e^{\lambda_1 t} = e^{-t/2}(\cos\omega t + j\sin\omega t)$:

$$2\,\mathrm{Re}\left[\tfrac{1}{2}\Big(\boldsymbol{I} + \tfrac{j}{\sqrt{7}}\boldsymbol{N}\Big)(\cos\omega t + j\sin\omega t)\right]e^{-t/2} = e^{-t/2}\Big[\cos(\omega t)\,\boldsymbol{I} - \frac{\sin(\omega t)}{\sqrt{7}}\boldsymbol{N}\Big]$$

(the real part of $(1+jb)(c+js)$ is $c-bs$). Writing out $-\boldsymbol{N}/\sqrt{7}$:

$$\boxed{\boldsymbol{\Phi}(t,0) = e^{\boldsymbol{A}t} = e^{-t/2}\begin{bmatrix} \cos\omega t + \frac{1}{\sqrt{7}}\sin\omega t & -\frac{4}{\sqrt{7}}\sin\omega t \\ \frac{2}{\sqrt{7}}\sin\omega t & \cos\omega t - \frac{1}{\sqrt{7}}\sin\omega t \end{bmatrix}, \qquad \omega = \frac{\sqrt{7}}{2}}$$

Checks.

- $t=0$: $\cos 0 = 1$, $\sin 0 = 0 \Rightarrow \boldsymbol{\Phi}(0,0) = \boldsymbol{I}$ ✓.
- Shortcut of Concepts §7.4 with $\sigma = -\frac{1}{2}$, $\omega = \frac{\sqrt{7}}{2}$: $\frac{\boldsymbol{A}-\sigma\boldsymbol{I}}{\omega} = \frac{2\boldsymbol{A}+\boldsymbol{I}}{\sqrt{7}} = \frac{1}{\sqrt{7}}\begin{bmatrix}1 & -4\\ 2 & -1\end{bmatrix} = -\frac{\boldsymbol{N}}{\sqrt{7}}$ ✓ — same result.
- Derivative at $t=0$: $-\frac{1}{2}\boldsymbol{I} + \omega\cdot\frac{1}{\sqrt{7}}\begin{bmatrix}1 & -4\\ 2 & -1\end{bmatrix} = -\frac{1}{2}\boldsymbol{I} + \frac{1}{2}\begin{bmatrix}1 & -4\\ 2 & -1\end{bmatrix} = \begin{bmatrix}0 & -2\\ 1 & -1\end{bmatrix} = \boldsymbol{A}$ ✓.

1(b) — Sum of modes; which mode diverges?

From Step 4, with $\boldsymbol{M}_1 = \boldsymbol{p}_1\tilde{\boldsymbol{p}}_1$ and $\boldsymbol{M}_2 = \boldsymbol{p}_2\tilde{\boldsymbol{p}}_2$:

$$\boxed{e^{\boldsymbol{A}t} = \underbrace{\frac{1}{2}\Big(\boldsymbol{I} + \frac{j}{\sqrt{7}}\begin{bmatrix}-1 & 4\\ -2 & 1\end{bmatrix}\Big)}_{\boldsymbol{p}_1\tilde{\boldsymbol{p}}_1} e^{\left(-\frac{1}{2}+j\frac{\sqrt{7}}{2}\right)t} + \underbrace{\frac{1}{2}\Big(\boldsymbol{I} - \frac{j}{\sqrt{7}}\begin{bmatrix}-1 & 4\\ -2 & 1\end{bmatrix}\Big)}_{\boldsymbol{p}_2\tilde{\boldsymbol{p}}_2} e^{\left(-\frac{1}{2}-j\frac{\sqrt{7}}{2}\right)t}}$$

Check: $\boldsymbol{M}_1 + \boldsymbol{M}_2 = \boldsymbol{I}$ ✓ (the value at $t=0$).

Which mode diverges? The size of each mode is set by the real part of its eigenvalue (Concepts §8.3):

$$|e^{\lambda_{1,2}t}| = e^{\mathrm{Re}(\lambda_{1,2})\,t} = e^{-t/2} \xrightarrow{t\to\infty} 0.$$

Neither mode diverges. Both eigenvalues have real part $-\frac{1}{2} < 0$, so both modes are decaying oscillations with frequency $\omega = \frac{\sqrt{7}}{2}$ and envelope $e^{-t/2}$ (time constant 2 s). $e^{\boldsymbol{A}t} \to 0$ for every initial state, and the system is **(asymptotically) stable** — there is no instability.

Remark. The question suggests that one mode diverges; for the matrix given on the sheet this is not the case. A mode would introduce divergence only if $\mathrm{Re}\,\lambda_i > 0$. Because the eigenvalues form a conjugate pair with the **same** real part, here both modes would always decay or diverge together — e.g. if the bottom-right entry were $+1$ instead of -1, the characteristic polynomial would be $\lambda^2 - \lambda + 2$ with $\mathrm{Re}\,\lambda = +\frac{1}{2}$, and both modes would diverge like $e^{t/2}$.

1(c) — Impulse response $u(t) = \delta(t)$

The initial state is zero, so the natural response vanishes and the **total response equals the forced response** (Concepts §12.2, $\boldsymbol{D} = 0$):

$$y(t) = \underbrace{\boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{x}(0)}_{=0} + \boldsymbol{C}\int_{0^-}^{t} e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\,\delta(\tau)\,\mathrm{d}\tau.$$

By the sifting property (Concepts §2.4) the integral picks out $\tau = 0$:

$$\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{B}, \qquad y(t) = \boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{B} = H(t) \qquad (t \geq 0).$$

State. $e^{\boldsymbol{A}t}\boldsymbol{B}$ = sum of the two columns of $e^{\boldsymbol{A}t}$ (because $\boldsymbol{B} = [1,1]^\top$):

$$\boldsymbol{x}(t) = e^{-t/2}\begin{bmatrix} \cos\omega t + \frac{1}{\sqrt{7}}\sin\omega t - \frac{4}{\sqrt{7}}\sin\omega t \\ \frac{2}{\sqrt{7}}\sin\omega t + \cos\omega t - \frac{1}{\sqrt{7}}\sin\omega t \end{bmatrix} = e^{-t/2}\begin{bmatrix} \cos\omega t - \frac{3}{\sqrt{7}}\sin\omega t \\ \cos\omega t + \frac{1}{\sqrt{7}}\sin\omega t \end{bmatrix}.$$

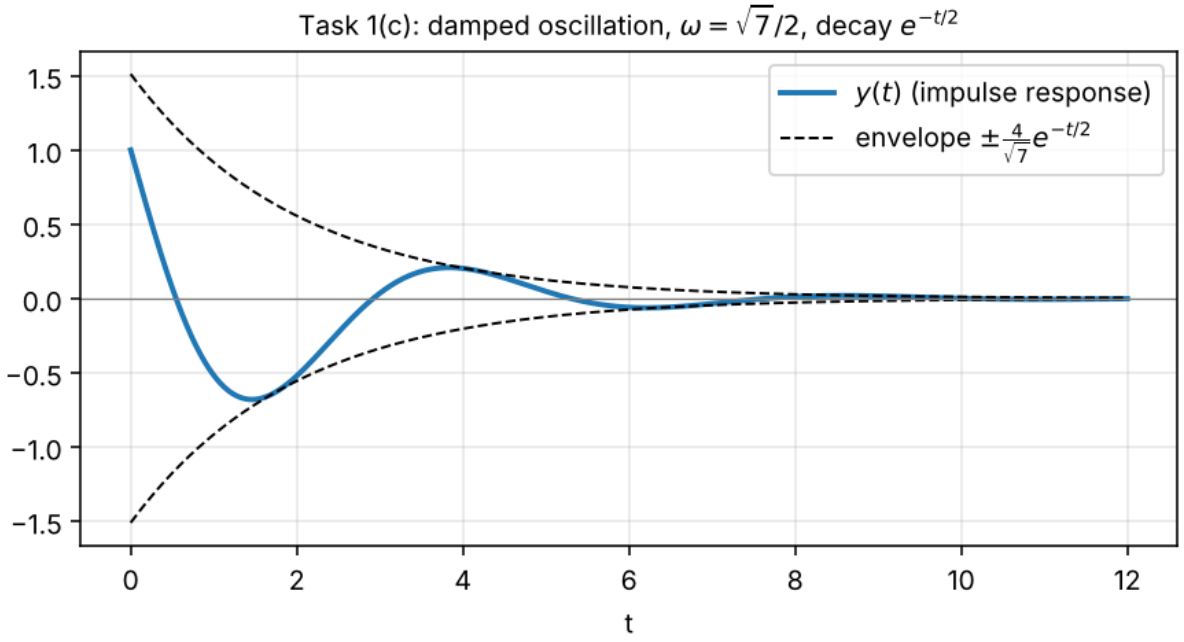
Output. $\boldsymbol{C} = [1\ 0]$ picks the first component:

$$\boxed{y(t) = \begin{cases} e^{-t/2}\Big(\cos\dfrac{\sqrt{7}}{2}t - \dfrac{3}{\sqrt{7}}\sin\dfrac{\sqrt{7}}{2}t\Big), & t \geq 0 \\ 0, & t < 0 \end{cases}}$$

Checks and interpretation.

- $y(0^+) = 1 = \boldsymbol{C}\boldsymbol{B}$ ✓: the impulse instantly sets the state to $\boldsymbol{x}(0^+) = \boldsymbol{B} = [1,1]^\top$ (Concepts §13.2), and the sensor sees $x_1 = 1$.
- After the hit the system rings at $\omega = \frac{\sqrt{7}}{2}$ rad/s with a decaying envelope. In amplitude-phase form $y(t) = \frac{4}{\sqrt{7}}e^{-t/2}\cos(\omega t + \varphi)$ with $\tan\varphi = \frac{3}{\sqrt{7}}$ (amplitude $\sqrt{1 + 9/7} = 4/\sqrt{7}$).
- $y(t) \to 0$, consistent with (b): the response to a hit dies out.

Check: `python/task1_modes_impulse.py` verifies $\boldsymbol{P}$, $\boldsymbol{P}^{-1}$, the mode matrices, $e^{\boldsymbol{A}t}$ against SymPy's exact matrix exponential, and $y(t)$ against `scipy.signal.impulse`.



Task 2 — Recovering the state transition matrix

Given two trajectories of $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$:

$$\boldsymbol{x}^{(1)}(t) = \begin{bmatrix} 4e^{-t} - 3e^{-2t} \\ -2e^{-t} + 3e^{-2t} \end{bmatrix} \text{ for } \boldsymbol{x}^{(1)}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{x}^{(2)}(t) = \begin{bmatrix} e^{-2t} \\ -e^{-2t} \end{bmatrix} \text{ for } \boldsymbol{x}^{(2)}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Idea (Concepts §4.3, §15.1): every trajectory satisfies $\boldsymbol{x}(t) = \boldsymbol{\Phi}(t,0)\boldsymbol{x}(0)$, so two trajectories with independent initial states determine $\boldsymbol{\Phi}$ completely.

Step 1 — Check the initial states. At $t = 0$: $\boldsymbol{x}^{(1)}(0) = [4-3,\ -2+3]^\top = [1,1]^\top$ ✓ and $\boldsymbol{x}^{(2)}(0) = [1,-1]^\top$ ✓. They are linearly independent (not multiples of each other).

Step 2 — Stack the trajectories.

$$\boldsymbol{X}(t) = [\boldsymbol{x}^{(1)}(t)\ \ \boldsymbol{x}^{(2)}(t)] = \begin{bmatrix} 4e^{-t} - 3e^{-2t} & e^{-2t} \\ -2e^{-t} + 3e^{-2t} & -e^{-2t} \end{bmatrix}, \qquad \boldsymbol{X}_0 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

and $\boldsymbol{X}(t) = \boldsymbol{\Phi}(t,0)\boldsymbol{X}_0$.

Step 3 — Invert $\boldsymbol{X}_0$. $\det \boldsymbol{X}_0 = -1 - 1 = -2$:

$$\boldsymbol{X}_0^{-1} = \frac{1}{-2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Step 4 — $\boldsymbol{\Phi}(t,0) = \boldsymbol{X}(t)\boldsymbol{X}_0^{-1}$. Multiplying by $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ means: first column = (col 1 + col 2)/2, second column = (col 1 − col 2)/2:

row 1: $\frac{1}{2}\big[(4e^{-t}-3e^{-2t})+e^{-2t}\big]=2e^{-t}-e^{-2t}$, $\quad\frac{1}{2}\big[(4e^{-t}-3e^{-2t})-e^{-2t}\big]=2e^{-t}-2e^{-2t}$

row 2: $\frac{1}{2}\big[(-2e^{-t}+3e^{-2t})-e^{-2t}\big]=-e^{-t}+e^{-2t}$, $\quad\frac{1}{2}\big[(-2e^{-t}+3e^{-2t})+e^{-2t}\big]=-e^{-t}+2e^{-2t}$

$$\boxed{\boldsymbol{\Phi}(t,0)=\begin{bmatrix}2e^{-t}-e^{-2t} & 2e^{-t}-2e^{-2t}\\ -e^{-t}+e^{-2t} & -e^{-t}+2e^{-2t}\end{bmatrix}}$$

Check: $\boldsymbol{\Phi}(0,0)=\begin{bmatrix}2-1 & 2-2\\ -1+1 & -1+2\end{bmatrix}=\boldsymbol{I}$ ✓.

Step 5 — $\boldsymbol{A}=\dot{\boldsymbol{\Phi}}(t,0)\big|_{t=0}$ (Definition 1.2 at $t=t_0$):

$$\dot{\boldsymbol{\Phi}}(t,0)=\begin{bmatrix}-2e^{-t}+2e^{-2t} & -2e^{-t}+4e^{-2t}\\ e^{-t}-2e^{-2t} & e^{-t}-4e^{-2t}\end{bmatrix}\xrightarrow{t=0}\boxed{\boldsymbol{A}=\begin{bmatrix}0 & 2\\ -1 & -3\end{bmatrix}}$$

Checks.

- Eigenvalues of $\boldsymbol{A}$: $\lambda^2+3\lambda+2=(\lambda+1)(\lambda+2)\Rightarrow -1,-2$ — exactly the exponentials e^{-t}, e^{-2t} in the data ✓.
- Eigenvector of -2: $(-2\boldsymbol{I}-\boldsymbol{A})\boldsymbol{v}=\begin{bmatrix}-2 & -2\\ 1 & 1\end{bmatrix}\boldsymbol{v}=0\Rightarrow\boldsymbol{v}=[1,-1]^\top$. That is $\boldsymbol{x}^{(2)}(0)$, which explains why the second trajectory contains **only** e^{-2t} — it is a pure mode (Concepts §8.2) ✓.
- Second trajectory: $\boldsymbol{A}\begin{bmatrix}e^{-2t}\\ -e^{-2t}\end{bmatrix}=\begin{bmatrix}-2e^{-2t}\\ -e^{-2t}+3e^{-2t}\end{bmatrix}=\begin{bmatrix}-2e^{-2t}\\ 2e^{-2t}\end{bmatrix}=\dot{\boldsymbol{x}}^{(2)}$ ✓.
- $\dot{\boldsymbol{\Phi}}=\boldsymbol{A}\boldsymbol{\Phi}$ for all t, and $e^{\boldsymbol{A}t}$ computed from $\boldsymbol{A}$ equals $\boldsymbol{\Phi}$ (script) ✓.

Check: `python/task2_recover_A.py` (symbolic), plus a simulation of $\dot{\boldsymbol{x}}=\boldsymbol{A}\boldsymbol{x}$ with the recovered $\boldsymbol{A}$ that reproduces both given trajectories.

Task 3 — Determining initial conditions

$$\boldsymbol{A}=\begin{bmatrix}-1 & 1 & 0\\ 0 & -1 & 0\\ 0 & 0 & 2\end{bmatrix},\qquad \boldsymbol{C}=\begin{bmatrix}1 & 1 & 1\end{bmatrix},\qquad \text{wanted: } \boldsymbol{x}(0) \text{ such that } y(t)=te^{-t},\ t\ge 0.$$

Approach: compute $y(t)=\boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{x}(0)$ for unknown $\boldsymbol{x}(0)$, then match coefficients (Concepts §15.2).

Step 1 — Structure of $\boldsymbol{A}$. $\boldsymbol{A}$ is block-diagonal: a 2×2 **Jordan block** for $\lambda=-1$ (upper-left) and a 1×1 block $\lambda=2$ (lower-right). It is already in Jordan form (Concepts §9), so exponentiate block by block (Concepts §5.2):

$$\exp\!\Big(\begin{bmatrix}-1 & 1\\ 0 & -1\end{bmatrix}t\Big)=e^{-t}\begin{bmatrix}1 & t\\ 0 & 1\end{bmatrix},\qquad e^{2t}\text{ for the last state.}$$

(The block is $-\boldsymbol{I}+\boldsymbol{N}$ with $\boldsymbol{N}=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$, $\boldsymbol{N}^2=0$, so $e^{(-\boldsymbol{I}+\boldsymbol{N})t}=e^{-t}(\boldsymbol{I}+\boldsymbol{N}t)$.)

$$e^{\boldsymbol{A}t}=\begin{bmatrix}e^{-t} & te^{-t} & 0\\ 0 & e^{-t} & 0\\ 0 & 0 & e^{2t}\end{bmatrix}.$$

Step 2 — Natural response for $\boldsymbol{x}(0) = [a, b, c]^\top$.

$$\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{x}(0) = \begin{bmatrix} ae^{-t} + bte^{-t} \\ be^{-t} \\ ce^{2t} \end{bmatrix}, \qquad y(t) = x_1 + x_2 + x_3 = (a+b)\,e^{-t} + b\,te^{-t} + c\,e^{2t}.$$

Step 3 — Match coefficients with $y(t) = 0 \cdot e^{-t} + 1 \cdot te^{-t} + 0 \cdot e^{2t}$. The functions e^{-t}, te^{-t}, e^{2t} are linearly independent (Concepts §2.3), so each coefficient must agree:

Function	Coefficient in $y(t)$	Required	Equation
te^{-t}	b	1	$b = 1$
e^{-t}	$a + b$	0	$a = -b = -1$
e^{2t}	c	0	$c = 0$

$$\boxed{\boldsymbol{x}(0) = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}$$

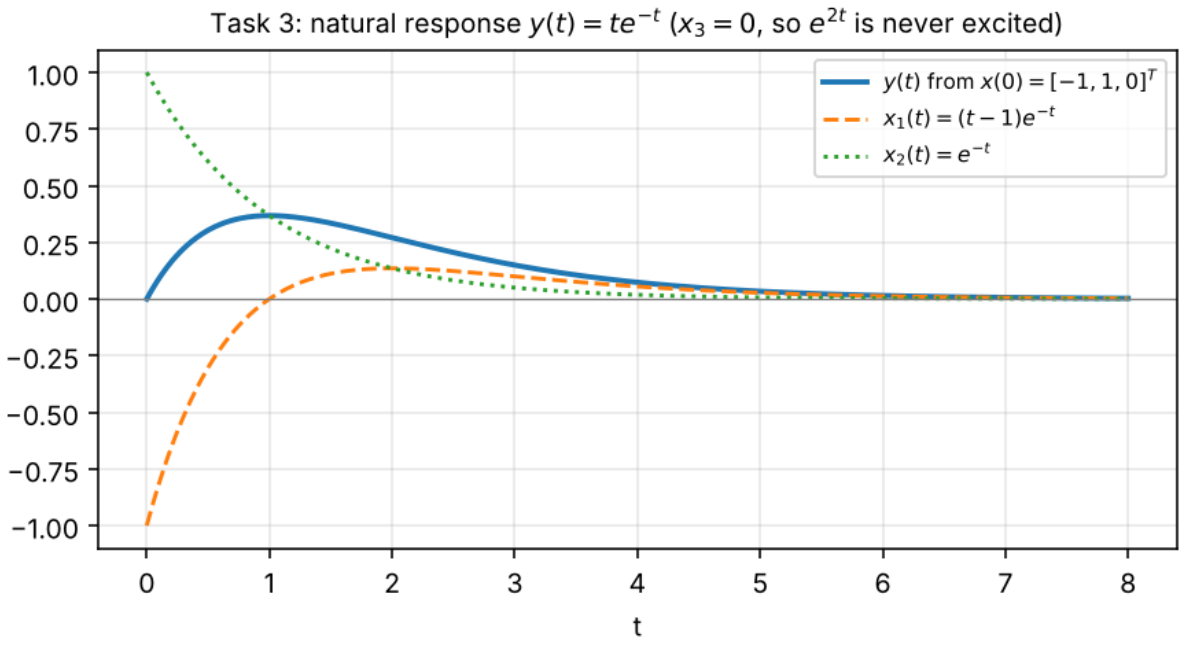
The solution is **unique** — three independent equations for three unknowns.

Check. $\boldsymbol{x}(t) = [(t-1)e^{-t},\ e^{-t},\ 0]^\top$, so $y(t) = (t-1)e^{-t} + e^{-t} + 0 = te^{-t}$ ✓.

Interpretation.

- A term te^{-t} can appear only because $\boldsymbol{A}$ has a **Jordan block** for $\lambda = -1$ (Concepts §9.2) — a diagonalizable $\boldsymbol{A}$ could never produce it.
- $c = 0$ is forced: x_3 carries the **unstable mode** e^{2t}, and since $\boldsymbol{C}$ sees x_3, any $c \neq 0$ would make $y(t)$ blow up. The chosen initial state does not excite the unstable mode — but the system itself is unstable, and the smallest error in $x_3(0)$ would eventually dominate.
- $a = -1$ cancels the e^{-t} that x_2 adds to the output, leaving only te^{-t}.

Check: `python/task3_initial_condition.py` (symbolic + simulation).



Task 4 — Cayley-Hamilton method and response of LTI systems

$$\dot{\boldsymbol{x}} = \underbrace{\begin{bmatrix} -6 & -1 \\ 5 & 0 \end{bmatrix}}_{\boldsymbol{A}} \boldsymbol{x} + \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\boldsymbol{B}} u, \qquad y = \underbrace{\begin{bmatrix} 1 & 1 \end{bmatrix}}_{\boldsymbol{C}} \boldsymbol{x}, \qquad \boldsymbol{x}(t_0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \ (t_0 > 0), \qquad u(t) = \sigma(t - t_0).$$

4(a) — $e^{\boldsymbol{A}t}$ with Cayley-Hamilton

Step 1 — Eigenvalues.

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \det \begin{bmatrix} \lambda + 6 & 1 \\ -5 & \lambda \end{bmatrix} = \lambda(\lambda+6)+5 = \lambda^2+6\lambda+5 = (\lambda+1)(\lambda+5) = 0 \implies \lambda_1 = -1, \quad \lambda_2 = -5.$$

Distinct, so the Vandermonde matrix is invertible (Concepts §10.2). (Cayley–Hamilton itself: $\boldsymbol{A}^2 + 6\boldsymbol{A} + 5\boldsymbol{I} = 0$, verified in Concepts §10.1.)

Step 2 — Ansatz. $n = 2$, so

$$e^{\boldsymbol{A}t} = \alpha_0(t)\boldsymbol{I} + \alpha_1(t)\boldsymbol{A}.$$

Step 3 — One scalar equation per eigenvalue ($e^{\lambda_i t} = \alpha_0 + \alpha_1 \lambda_i$):

$$\begin{bmatrix} e^{-t} \\ e^{-5t} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 \\ 1 & -5 \end{bmatrix}}_{\boldsymbol{V}} \begin{bmatrix} \alpha_0(t) \\ \alpha_1(t) \end{bmatrix} \quad \Longleftrightarrow \quad \begin{cases} e^{-t} = \alpha_0 - \alpha_1 \\ e^{-5t} = \alpha_0 - 5\alpha_1 \end{cases}$$

Step 4 — Solve. Subtract the second equation from the first: $e^{-t} - e^{-5t} = 4\alpha_1$, then $\alpha_0 = e^{-t} + \alpha_1$:

$$\boxed{\alpha_1(t) = \frac{e^{-t} - e^{-5t}}{4}, \qquad \alpha_0(t) = \frac{5e^{-t} - e^{-5t}}{4}}$$

(Same as the general 2×2 formula of Concepts §10.2.) *Check* $t = 0$*:* $\alpha_0 = 1$, $\alpha_1 = 0 \Rightarrow e^{\boldsymbol{A}\cdot 0} = \boldsymbol{I}$ ✓.

Step 5 — Insert.

$$e^{\boldsymbol{A}t} = \frac{5e^{-t} - e^{-5t}}{4}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} + \frac{e^{-t} - e^{-5t}}{4}\begin{bmatrix}-6 & -1\\ 5 & 0\end{bmatrix}.$$

Entry by entry (everything over 4):

- (1,1): $5e^{-t} - e^{-5t} - 6e^{-t} + 6e^{-5t} = -e^{-t} + 5e^{-5t}$
- (1,2): $-(e^{-t} - e^{-5t}) = -e^{-t} + e^{-5t}$
- (2,1): $5(e^{-t} - e^{-5t}) = 5e^{-t} - 5e^{-5t}$
- (2,2): $5e^{-t} - e^{-5t} + 0 = 5e^{-t} - e^{-5t}$

$$\boxed{e^{\boldsymbol{A}t} = \frac{1}{4}\begin{bmatrix}-e^{-t} + 5e^{-5t} & -e^{-t} + e^{-5t}\\ 5e^{-t} - 5e^{-5t} & 5e^{-t} - e^{-5t}\end{bmatrix}}$$

Checks: $t = 0$: $\frac{1}{4}\begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix} = \boldsymbol{I}$ ✓. Derivative at $t = 0$: $\frac{1}{4}\begin{bmatrix}1-25 & 1-5\\ -5+25 & -5+5\end{bmatrix} = \begin{bmatrix}-6 & -1\\ 5 & 0\end{bmatrix} = \boldsymbol{A}$ ✓.

4(b) — Response $y(t)$ for $t \geq 0$

The state is given at $t_0 > 0$ and the step switches on at t_0. With $\boldsymbol{\Phi}(t, t_0) = e^{\boldsymbol{A}(t-t_0)}$ the total response is (Concepts §12.2)

$$y(t) = \boldsymbol{C}e^{\boldsymbol{A}(t-t_0)}\boldsymbol{x}(t_0) + \boldsymbol{C}\int_{t_0}^{t} e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\,\sigma(\tau - t_0)\,\mathrm{d}\tau.$$

Write $\tau_0 := t - t_0$ for the time elapsed since t_0 (negative for $t < t_0$).

Step 1 — Natural part (valid for all t).

$$e^{\boldsymbol{A}\tau_0}\boldsymbol{x}(t_0) = \frac{1}{4}\begin{bmatrix}(-e^{-\tau_0} + 5e^{-5\tau_0}) + 2(-e^{-\tau_0} + e^{-5\tau_0})\\ (5e^{-\tau_0} - 5e^{-5\tau_0}) + 2(5e^{-\tau_0} - e^{-5\tau_0})\end{bmatrix} = \frac{1}{4}\begin{bmatrix}-3e^{-\tau_0} + 7e^{-5\tau_0}\\ 15e^{-\tau_0} - 7e^{-5\tau_0}\end{bmatrix},$$

$$y_h(t) = \boldsymbol{C}e^{\boldsymbol{A}\tau_0}\boldsymbol{x}(t_0) = \frac{1}{4}(12e^{-\tau_0} + 0\cdot e^{-5\tau_0}) = 3e^{-(t-t_0)}.$$

The e^{-5t} terms **cancel in the output**. Reason: the eigenvector of $\lambda_2 = -5$ is $\boldsymbol{v}_2 = [1, -1]^\top$ (from $(-5\boldsymbol{I} - \boldsymbol{A})\boldsymbol{v} = \begin{bmatrix}1 & 1\\ -5 & -5\end{bmatrix}\boldsymbol{v} = 0$), and $\boldsymbol{C}\boldsymbol{v}_2 = 1 - 1 = 0$: the sensor cannot see the fast mode at all. (Week 12 calls this *unobservable*.)

Step 2 — Case $0 \le t \le t_0$. Here $\sigma(t-t_0) = 0$ — no input yet, and the forced integral is empty. The state is known at t_0, so it is propagated **backwards** with $\boldsymbol{\Phi}(t,t_0) = e^{\boldsymbol{A}(t-t_0)}$, $t - t_0 \le 0$ (Concepts §4.2, $\boldsymbol{\Phi}(t,t_0)^{-1} = \boldsymbol{\Phi}(t_0,t)$):

$$\boxed{y(t) = 3e^{-(t-t_0)} = 3e^{t_0-t}, \qquad 0 \le t \le t_0}$$

At $t = 0$: $y(0) = 3e^{t_0}$. The output was **larger** in the past and has been decaying towards the value 3 that it reaches at t_0.

Step 3 — Forced part for $t \ge t_0$. For $\tau \ge t_0$ the input is 1, so (Concepts §13.3, $\boldsymbol{A}$ invertible since $\det \boldsymbol{A} = 5 \ne 0$):

$$\boldsymbol{x}_p(t) = \int_{t_0}^{t} e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\,\mathrm{d}\tau = \boldsymbol{A}^{-1}(e^{\boldsymbol{A}\tau_0} - \boldsymbol{I})\boldsymbol{B}, \qquad \boldsymbol{A}^{-1} = \frac{1}{5}\begin{bmatrix} 0 & 1 \\ -5 & -6 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{5} \\ -1 & -\frac{6}{5} \end{bmatrix}.$$

- $\boldsymbol{A}^{-1}\boldsymbol{B} = [0,\ -1]^\top$ (first column of $\boldsymbol{A}^{-1}$).
- $e^{\boldsymbol{A}\tau_0}\boldsymbol{B}$ = first column of $e^{\boldsymbol{A}\tau_0} = \frac{1}{4}[-e^{-\tau_0} + 5e^{-5\tau_0},\ 5e^{-\tau_0} - 5e^{-5\tau_0}]^\top$.
- $\boldsymbol{A}^{-1}e^{\boldsymbol{A}\tau_0}\boldsymbol{B}$: first row $\frac{1}{5}\cdot\frac{1}{4}(5e^{-\tau_0} - 5e^{-5\tau_0}) = \frac{1}{4}(e^{-\tau_0} - e^{-5\tau_0})$; second row $\frac{1}{4}\big[(e^{-\tau_0} - 5e^{-5\tau_0}) - \frac{6}{5}(5e^{-\tau_0} - 5e^{-5\tau_0})\big] = \frac{1}{4}(-5e^{-\tau_0} + e^{-5\tau_0})$.

$$\boldsymbol{x}_p(t) = \frac{1}{4}\begin{bmatrix} e^{-\tau_0} - e^{-5\tau_0} \\ -5e^{-\tau_0} + e^{-5\tau_0} \end{bmatrix} - \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{4}(e^{-\tau_0} - e^{-5\tau_0}) \\ 1 + \frac{1}{4}(-5e^{-\tau_0} + e^{-5\tau_0}) \end{bmatrix}, \qquad y_p(t) = \boldsymbol{C}\boldsymbol{x}_p = 1 - e^{-(t-t_0)}.$$

Again the $e^{-5\tau_0}$ terms cancel in y.

Step 4 — Total response for $t \ge t_0$.

$$\boxed{y(t) = \underbrace{3e^{-(t-t_0)}}_{\text{natural}} + \underbrace{1 - e^{-(t-t_0)}}_{\text{forced}} = 1 + 2e^{-(t-t_0)}, \qquad t \ge t_0}$$

For completeness, the state: $\boldsymbol{x}(t) = \big[-\frac{1}{2}e^{-\tau_0} + \frac{3}{2}e^{-5\tau_0},\ \ 1 + \frac{5}{2}e^{-\tau_0} - \frac{3}{2}e^{-5\tau_0}\big]^\top$ ($t \ge t_0$), which gives $[1,2]^\top$ at $\tau_0 = 0$ ✓.

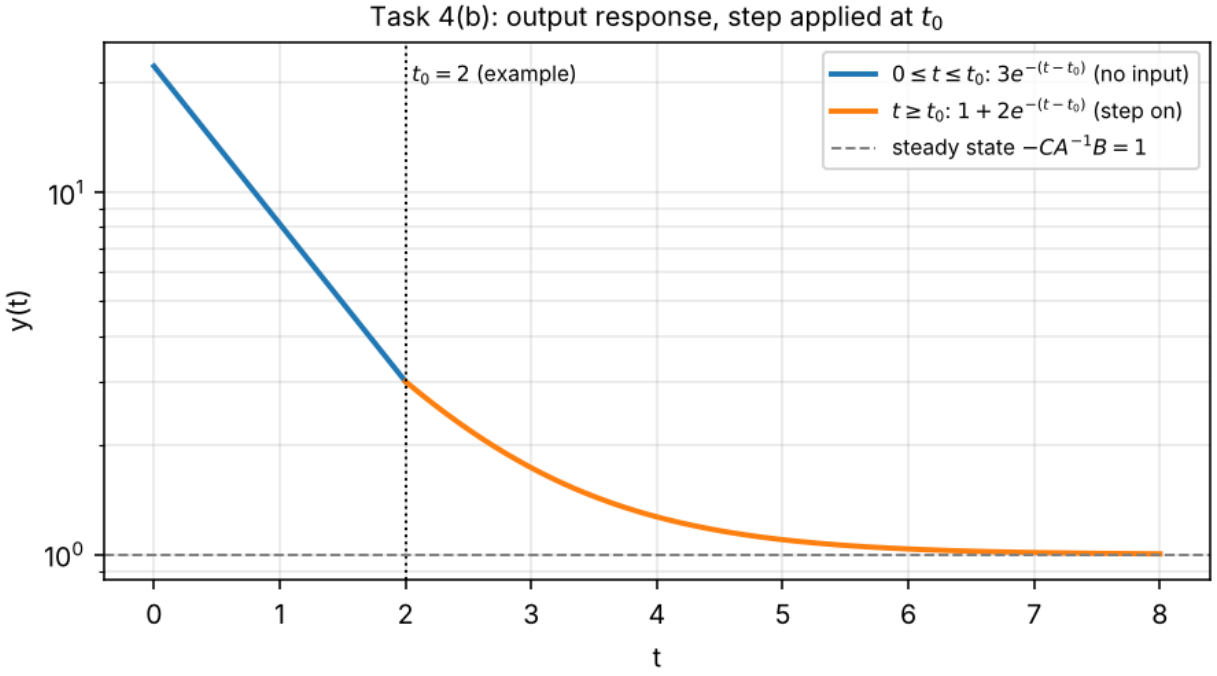
Summary of (b):

$$y(t) = \begin{cases} 3e^{-(t-t_0)}, & 0 \le t \le t_0 \\ 1 + 2e^{-(t-t_0)}, & t \ge t_0 \end{cases}$$

Checks.

- **Continuity at t_0:** both branches give $y(t_0) = 3 = \boldsymbol{C}\boldsymbol{x}(t_0) = 1 + 2$ ✓ (no feedthrough, $\boldsymbol{D} = 0$, so a step cannot make y jump).
- **Steady state:** $y(\infty) = -\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B} = -[1\ 1][0,-1]^\top = 1$ ✓ — both eigenvalues are negative, so the response settles.
- **Time invariance:** the result depends only on $t - t_0$ — the step response at t_0 is the step response at 0 shifted by t_0.

Check: `python/task4_cayley_hamilton.py` verifies the α_k, $e^{\boldsymbol{A}t}$, both branches symbolically, and simulates the system backwards from t_0 to 0 and forwards from t_0 (example $t_0 = 2$):



Summary of results

- **Task 1:** $\lambda_{1,2} = -\frac{1}{2} \pm j\frac{\sqrt{7}}{2}$; $e^{\boldsymbol{A}t} = e^{-t/2}\begin{bmatrix} \cos\omega t + \frac{1}{\sqrt{7}}\sin\omega t & -\frac{4}{\sqrt{7}}\sin\omega t \\ \frac{2}{\sqrt{7}}\sin\omega t & \cos\omega t - \frac{1}{\sqrt{7}}\sin\omega t \end{bmatrix}$, $\omega = \frac{\sqrt{7}}{2}$; modes $\boldsymbol{p}_i\tilde{\boldsymbol{p}}_i e^{\lambda_i t}$ with $\boldsymbol{p}_1\tilde{\boldsymbol{p}}_1 = \frac{1}{2}(\boldsymbol{I} + \frac{j}{\sqrt{7}}\boldsymbol{N})$; **no mode diverges** (Re $\lambda = -\frac{1}{2}$); impulse response $y(t) = e^{-t/2}\big(\cos\omega t - \frac{3}{\sqrt{7}}\sin\omega t\big)$, $t \ge 0$.
- **Task 2:** $\boldsymbol{\Phi}(t,0) = \begin{bmatrix} 2e^{-t} - e^{-2t} & 2e^{-t} - 2e^{-2t} \\ -e^{-t} + e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$, $\boldsymbol{A} = \begin{bmatrix} 0 & 2 \\ -1 & -3 \end{bmatrix}$.
- **Task 3:** $\boldsymbol{x}(0) = [-1,\ 1,\ 0]^\top$ (unique).
- **Task 4:** $\alpha_0 = \frac{5e^{-t} - e^{-5t}}{4}$, $\alpha_1 = \frac{e^{-t} - e^{-5t}}{4}$, $e^{\boldsymbol{A}t} = \frac{1}{4}\begin{bmatrix} -e^{-t} + 5e^{-5t} & -e^{-t} + e^{-5t} \\ 5e^{-t} - 5e^{-5t} & 5e^{-t} - e^{-5t} \end{bmatrix}$; $y(t) = 3e^{-(t-t_0)}$ for $0 \le t \le t_0$, $y(t) = 1 + 2e^{-(t-t_0)}$ for $t \ge t_0$.